EmoCare: A Multimodal AI Framework for Emotional Wellbeing Screening and Support

System / repository name: EmoCare

Submitted in partial fulfillment of the requirements of the degree of

Bachelor of Technology in Electronics and Telecommunication Engineering

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PROJECT REPORT APPROVAL FOR B.TECH

This project report titled

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is approved for the degree of

Undergraduate

in

"Bachelor of Technology in Electronics and Telecommunication Engineering"

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Date: 24 October 2026 Place: Mumbai

CERTIFICATE OF APPROVAL

This is to certify that the project entitled

"EmoCare: A Multimodal AI Framework for Emotional Wellbeing Screening and Support"

System / repository name: EmoCare

is a bonafide work of

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Date: 24 October 2026

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DECLARATION

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented, fabricated or falsified any idea, data, fact or source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have not been properly cited or from whom proper permission has not been taken when needed.

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ABSTRACT

Emotional distress is common among young adults, yet signs may appear in everyday language, voice and behaviour rather than in clinical settings. EmoCare is a web-based emotional-wellbeing prototype combining text analysis, speech analysis, self-reported behavioural metrics and the PHQ-9 and GAD-7 screening questionnaires. It produces Low, Moderate or High risk indications and supportive guidance. These outputs are not clinical diagnoses.

The source repository uses a Next.js 16 / React 19 frontend and a FastAPI backend.

Execution evidence from commit f0532e6 confirms that all four model modules loaded, 13 REST routes were documented, eight frontend routes rendered with placeholder Supabase configuration, and the journal text-to-result path was exercised end to end. Saved artefacts report text accuracy 0.745 (macro F1 0.481; High-tier recall 0.262) and speech emotion accuracy 0.507 (macro F1 0.481). The 1.0 fusion score derives from synthetic probability vectors and is not valid evidence of real-world performance. The merged commit and its tests were inspected at source level but not re-executed.

Ten limitations are documented for the final version (Table 5.6). The most important are: crisis-safety handling and the new assessment and therapy tests have not been executed; the behavioural endpoint maps seven user inputs to a 50-feature model with zero-padding; speech feature-name alignment against the saved 95-column schema is not verified; the fusion model was trained on synthetic probabilities; text High-tier recall is low; and raw training data and licences are unavailable for reproduction. EmoCare remains a non-diagnostic prototype.

Keywords: emotion recognition, multimodal fusion, speech emotion recognition, text classification, mental-health screening, FastAPI, Next.js, responsible AI.

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LIST OF ABBREVIATIONS AND SYMBOLS

Abbreviation	Expansion
AI	Artificial Intelligence
API	Application Programming Interface
CBT	Cognitive Behavioural Therapy
CORS	Cross-Origin Resource Sharing
CSV	Comma-Separated Values
EDA	Exploratory Data Analysis
F0	Fundamental frequency (pitch)
GAD-7	Generalized Anxiety Disorder 7-item scale
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
LLM	Large Language Model
LR	Logistic Regression
MFCC	Mel-Frequency Cepstral Coefficients
ML	Machine Learning
NLP	Natural Language Processing
PHQ-9	Patient Health Questionnaire 9-item
REST	Representational State Transfer
RF	Random Forest
SER	Speech Emotion Recognition
STT / TTS	Speech-to-Text / Text-to-Speech
TF-IDF	Term Frequency - Inverse Document Frequency
XGBoost	Extreme Gradient Boosting

Symbols

Symbol	Meaning	Unit
t(w, d)	TF-IDF weight of term w in document d	dimensionless
p = (pL, pM, pH)	Probability vector over Low, Moderate, High tiers	dimensionless
A	Cross-modal agreement score	0 to 1
sr	Audio sampling rate (22,050 Hz in the code)	Hz
F1	Harmonic mean of precision and recall	0 to 1
theta_H	Escalation threshold on high-risk probability (0.35)	dimensionless

List of Publications: Not applicable. No publication or conference acceptance related to this work has been confirmed.

CHAPTER 1: INTRODUCTION

1.1 Introduction to EmoCare

With EmoCare, a user can write a journal entry, record a voice note, log daily habits (sleep, activity, social contact, stress, mood), fill in the PHQ-9 and GAD-7 questionnaires, and chat with an AI companion. Separate ML models rate each input as Low, Moderate or High emotional risk. A fusion model then combines those ratings into a single result, along with an urgency level and some rule-based coping suggestions.

EmoCare is a screening and reflection tool, not a diagnostic one. It can't diagnose depression, anxiety or anything else, and its outputs shouldn't be read as a clinical assessment. This report covers the system as I found it and ran it in the repository, limitations included.

1.2 Background of the Problem

Emotional distress is common among students and young adults, and many people don't seek help until symptoms are well advanced. The early signs often show up in everyday behaviour: how someone writes, how their voice sounds, and shifts in their sleep, activity and social contact. Self-report questionnaires like the PHQ-9 [1] and GAD-7 [2] are widely used to screen for depression and anxiety, but they're only given now and then, and they rely on the person being willing to fill them in.

Affective computing tries to recognise emotional states automatically from signals like text and speech. Classical emotion theory describes a small set of basic emotions [3], and datasets built around those categories have made machine-learning approaches to emotion recognition possible. Using several modalities together makes sense because any single channel can be ambiguous or missing.

1.3 Problem Statement

The aim was to build a web app that picks up emotional signals from text, speech and self-reported behaviour, and estimates a risk level for each one. It then combines those into a single result that users can see the reasoning behind, with a flag that escalates when the models are unsure. The app responds with safe, supportive guidance and makes clear throughout that it isn't a diagnostic tool.

1.4 Motivation

- Low-effort reflection tools could help people notice sustained distress before it gets worse.
- A single-channel system breaks down when that channel is missing or contradicts itself. Fusing several channels lets the system show whether they agree.
- Mental-health apps have safety responsibilities, so a project that classifies risk has to look openly at how it can fail.
- The project covers signal processing (audio features), machine learning, full-stack web development and responsible AI, which suits an Electronics and Telecommunication project spanning signals, systems and software.

1.5 Aim and Objectives

Aim: to build and test a multimodal emotional-risk screening prototype, and to state plainly what it does and doesn't achieve.

- O1. Collect text, voice and behavioural inputs through a usable web interface.
- O2. Extract text features (TF-IDF, linguistic markers) and audio features (MFCCs, spectral descriptors), then train classifiers on them.
- O3. Build a late-fusion engine and an escalation policy that use risk probability and agreement between modalities.
- O4. Implement PHQ-9 and GAD-7 scoring with plain-language feedback.
- O5. Provide a supportive conversation module with a deterministic crisis-handling path.
- O6. Check that the working system runs, evaluate it with the artefacts available, and report its limitations.

1.6 Scope of the Project

In scope are the web frontend, the FastAPI inference backend, the saved model artefacts, the questionnaire module, the conversation module and local persistence. Out of scope are clinical validation, diagnosis, deployment to production, and any claim of clinical accuracy. Datasets named in the repository README (GoEmotions, Dreaddit, RAVDESS, CREMA-D, DAIC-WOZ, IEMOCAP, StudentLife) were checked individually; Section 3.7 states which were actually used.

1.7 Proposed Solution

The solution is a three-tier system: a Next.js frontend; a FastAPI backend that loads joblib model artefacts once at start-up and exposes REST endpoints; and a persistence layer (SQLite by default, PostgreSQL optionally). Each modality produces a probability vector over the three risk tiers, and an XGBoost meta-classifier fuses the three vectors. A deterministic coping-advice library supplies suggestions, and a safety gate is intended to bypass the language model when high risk is detected.

1.8 Applications and Intended Users

- Students and young adults seeking a private reflective journaling and check-in tool.
- Campus wellbeing teams exploring prototype screening workflows (with human review, not automated decisions).
- Researchers and students studying multimodal affective computing and responsible-AI design.

1.9 Organization of the Report

- Chapter 2 reviews literature on text, speech and multimodal emotion recognition and identifies the research gap.
- Chapter 3 presents the analysis, architecture, datasets, models and workflows.
- Chapter 4 describes the implementation, repository, interfaces and execution procedure with real screenshots.
- Chapter 5 reports testing, available results, and failure cases.
- Chapter 6 concludes and proposes future work.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction to Emotion Recognition

Emotion recognition is the computational estimation of affective states from observable signals. Poria et al. [4] review the progression from unimodal analysis to multimodal fusion and discuss feature-level and decision-level strategies. This project uses decision-level (late) fusion, in which each modality is classified separately and the resulting probability vectors are combined.

2.2 Text-Based Emotion Recognition

Large text corpora have driven progress in text emotion recognition. GoEmotions [5] provides about 58,000 Reddit comments annotated with 27 emotion categories or neutral,

Currently, the common route to high-performance models for emotion detection often involves the use of pre-trained transformers such as BERT [7] and domain-adapted variants such as MentalBERT [8]. However, approaches based on simpler features such as TF-IDF and linear models remain attractive due to the speed and inspectability of the inferences made. Such is the approach that has been deployed in EmoCare.

Research in the field of psychology has linked the style of language used by individuals to their emotional state. For instance, Rude et al. [9] found that students who were depressed used more first-person singular pronouns in their language. Similarly, Al-Mosaiwi and Johnstone [10] reported that individuals in anxiety, depression, and suicidal-ideation communities used elevated levels of absolutist language. EmoCare computes markers based on these findings, but these markers are only used as descriptive indicators of the emotional state of the individual.

2.3 Speech Emotion Recognition

Speech emotion recognition (SER) estimates emotion from the acoustic signal. Acted-emotion corpora such as RAVDESS [11] and CREMA-D [12] and the multimodal IEMOCAP corpus [13] are commonly used. Hand-crafted features such as Mel-Frequency Cepstral Coefficients (MFCC), pitch, energy and spectral descriptors, extracted with libraries such as librosa [14], feed classical classifiers, while self-supervised networks such as wav2vec 2.0 [15] learn representations directly from audio. EmoCare uses the hand-crafted route.

2.4 Multimodal Emotion Recognition and Mental-Health Data

Combining modalities can improve robustness when one channel is missing or noisy. Passive behavioural sensing has also been studied; the StudentLife study [16] related

smartphone-derived behaviour of college students to mental-health outcomes. Clinical interview corpora such as DAIC-WOZ [17] combine speech, video and transcripts with depression-related labels but are access-restricted. In this repository DAIC-WOZ and IEMOCAP appear only as folders in the README; no code or notebook uses them.

2.5 Relevant Machine-Learning Methods

Random Forest [18] integrates the predictions of numerous decision trees trained on different bootstrap samples of the same data set; gradient boosting as implemented in XGBoost [19] adds trees in sequence to minimize the residual error of the previous ensemble; both of these algorithms are available through widely used libraries such as scikit-learn [20].

2.6 Relevant Datasets and Existing Research

Section 3.7 lists datasets with their verified roles in this project. The literature shows that acted-speech corpora and social-media text corpora are convenient but differ from real conversational, self-reported distress, which limits generalisation.

2.7 Comparative Literature Survey

Study	Topic	Key finding	Limitation	Relevance to EmoCare
Poria et al., 2017 [4]	Review of affective computing from unimodal to multimodal fusion	Surveys feature extraction and fusion strategies across text, audio and video	A review; does not provide a deployable system or mental-health-specific validation	Motivates the fusion design
Demszky et al., 2020 [5]	GoEmotions: 58k Reddit comments labelled with 27 emotions or neutral	Introduces a large fine-grained emotion corpus; a BERT baseline reached an average F1 of 0.46	Reddit text; fine-grained labels remain hard to predict	Named in README; used only for EDA in this repository
Turcan and McKeown, 2019 [6]	Dreaddit: Reddit stress corpus	190K posts from five community categories; 3.5K segments labelled via crowdsourcing	Stress, not clinical diagnosis; social-media domain	Named in README; used only for EDA here
Devlin et al., 2019 [7]	BERT bidirectional transformer pre-training	Fine-tuning pre-trained transformers improved many language tasks	Compute-intensive	Not implemented in the deployed text model
Ji et al., 2022 [8]	MentalBERT: language models pre-trained for mental-health text	Domain-specific pre-training for mental-health classification	Requires large models and careful ethics	Referenced in repository docs; the deployed model is TF-IDF + LR
Rude et al., 2004 [9]	Language use of depressed and depression-vulnerable students	Depressed writers used more first-person singular pronouns	Small student sample; essays	Basis for the first-person marker
Al-Mosaiwi and Johnstone, 2018 [10]	Absolutist words in anxiety, depression and suicidal ideation	Absolutist words were more frequent in these groups	Group-level, not individual-level, evidence	Basis for the absolutist marker
Livingstone and Russo, 2018 [11]	RAVDESS validated emotional speech and song database	24 actors; 7,356 recordings; speech emotions include calm, happy, sad, angry, fearful, surprise and disgust	Acted emotion in a neutral accent	EDA notebook and speech modelling
Cao et al., 2014 [12]	CREMA-D crowd-sourced emotional actors dataset	Actors expressing emotions in sentence-level speech with crowd ratings	Acted speech	EDA notebook; speech modelling
Wang et al., 2014 [16]	StudentLife smartphone sensing study	Behavioural sensing of college students linked to mental-health measures	Small cohort; sensor features differ from self-report	Behavioural features and model

Table 2.1: Comparative survey of the literature relevant to EmoCare

2.8 Research Gap

Existing work provides datasets and strong single-modality models, but integrated, transparent, student-oriented prototypes that combine text, voice, self-reported behaviour and validated questionnaires, and that attach an explicit uncertainty and escalation policy, are less commonly documented in an end-to-end, reproducible form. EmoCare is an engineering integration of established techniques rather than a novel model: it does not introduce a new algorithm and makes no claim of solving an open research problem. A further gap, which this project helps expose, is that integrated systems need honest evaluation of safety behaviour, not only aggregate accuracy.

2.9 Summary

Literature supports the use of language style, acoustic features and behavioural signals as emotional-distress indicators, as well as fusion at the decision level as a practical combination strategy. The evidence that has been cited is at the group level and within specific domains, so predictions made for an individual must be treated as uncertain.

CHAPTER 3: SYSTEM ANALYSIS, DESIGN AND METHODOLOGY

3.1 System Overview

The system known as EmoCare functions through a client-server architecture. The Next.js browser client collects inputs and displays results; the FastAPI server validates requests, computes features, loads model artefacts, applies scoring and escalation logic, and persists assessments and therapy sessions. The last merge introduced a multi-level PHQ-9/GAD-7 process which includes consent collection and review stages and follow-up based on context and safety modal display and assessment results presentation and chat/voice session management with intent detection functionality. The code underwent inspection during commit cf39c66 but its runtime behavior did not receive any re-execution for this particular report.

3.2 Existing Approaches and Their Limitations

- Single-modality classifiers cannot express disagreement between channels.
- Periodic questionnaires alone are infrequent and effortful.
- Unconstrained generative chatbots may produce unsafe or unverified advice; EmoCare therefore constrains the language model to a catalogue of predefined coping tools.

3.3 Proposed System

The proposed system pairs per-modality classifiers with a late-fusion meta-classifier, an agreement score and a threshold-based escalation policy, and adds validated questionnaires and a constrained conversational layer.

3.4 Functional Requirements

ID	Requirement
FR1	Accept free-text journal entries and return a risk tier, probabilities and linguistic markers
FR2	Accept an audio upload and return emotion, risk tier and prosodic summary
FR3	Accept daily behavioural metrics and return a risk tier
FR4	Fuse any available modalities and return tier, urgency, agreement and coping advice
FR5	Serve PHQ-9 and GAD-7 questionnaires and score submissions
FR6	Manage chat and voice therapy sessions with stored history
FR7	Show crisis helpline information on high-risk or critical outcomes

Table 3.1: Functional requirements

3.5 Non-Functional Requirements

Attribute	Requirement
Usability	Simple responsive interface with plain-language feedback
Performance	Models loaded once at start-up; text inference runs locally
Portability	Runs on Python 3.12 / Node 22 locally; Docker files provided but not verified here
Privacy	Local SQLite default; third-party services only when keys are configured
Safety	Not diagnostic; crisis resources displayed; see limitations
Maintainability	Separate frontend, API, inference and persistence modules

Table 3.2: Non-functional requirements

3.6 Overall Architecture

Figure 3.1 shows the source-level architecture. The frontend pages call the REST API through a thin client module (frontend/src/lib/api.ts). The backend (backend/app/main.py) delegates all prediction tasks to inference.py and uses assessment.py for questionnaire scoring and therapy.py together with the engine/ package to handle conversations and stores data through db.py. The system allows users to connect with external services which include Supabase authentication and language-model providers and speech-to-text and text-to-speech providers although users must supply credentials which remain outside the repository.

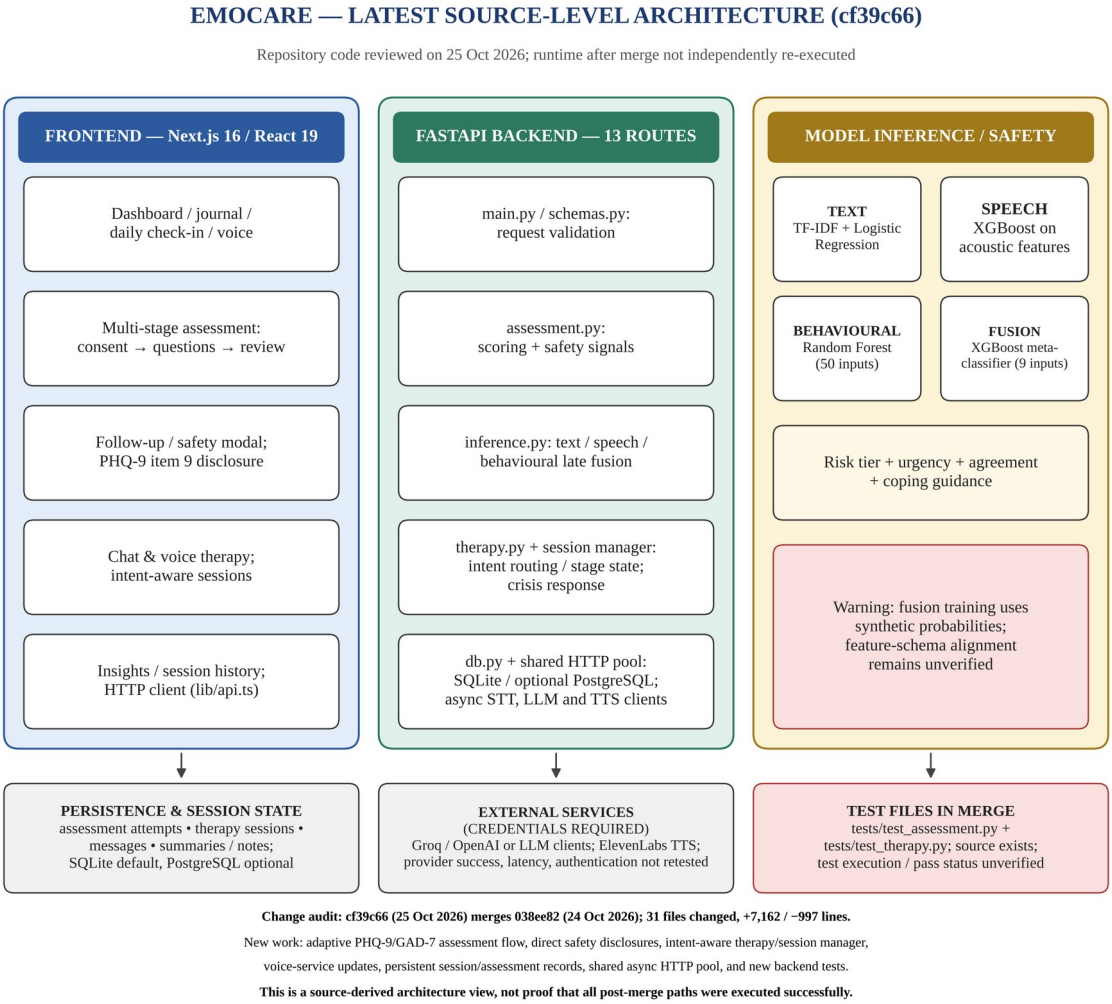


Figure 3.1: Updated source-level architecture after commit cf39c66 (runtime not re-verified)

3.7 Dataset Description

Table 3.3 separates datasets named in repository documentation from those actually evidenced in notebooks or model artefacts. Raw and processed training datasets were not available in the inspected materials, so model training and the saved metrics could not be reproduced.

Dataset	Evidence found	Role in the deployed system
GoEmotions [5]	EDA notebook only	Not used in any deployed model
Dreaddit [6]	EDA notebook only	Not used in any deployed model
Depression-severity text data	EDA notebook; text-model training (train/val/test CSVs)	Used. Original source and licence not stated in the repository: Not verified
RAVDESS [11]	EDA notebook; speech features	Used (combined speech feature table). Exact per-source composition: Not verified
CREMA-D [12]	EDA notebook; speech features	Used (combined speech feature table). Exact per-source composition: Not verified
StudentLife [16]	EDA notebook; behavioural features CSV; Random Forest	Used
DAIC-WOZ [17], IEMOCAP [13]	README folder names only	Not used: no notebook or code references

Table 3.3: Datasets named in the README and their verified usage

3.8 Data Preprocessing

Text. The text model was trained from pre-cleaned CSV files (a text_clean column when present) with rows lacking text or label removed. **Speech.** Acoustic features are stored in a combined feature table with train, validation and test splits already assigned. Whether the split separates speakers is Not verified. **Behavioural.** The training table is a StudentLife-derived feature file; missing values are filled by a median imputer and values are standardised.

3.9 Feature Extraction

3.9.1 Text features

The deployed vectoriser uses TF-IDF with unigrams to trigrams, sublinear term-frequency scaling, English stop word removal and a vocabulary of 10,000 terms. The weight of a term w in document d is calculated as follows:

$$t(w, d) = (1 + \ln tf(w, d)) \times idf(w) \quad (3.1)$$

Three descriptive linguistic markers are also computed at request time (Table 3.4). They are returned to the interface but are not inputs to the classifier.

Marker	Word list (abridged)	Definition
First-person rate	i, me, my, mine, myself, we, us, our, ours	count / words
Absolutist rate	always, never, completely, nothing, everything, forever, ...	count / words
Negation rate	not, no, never, cannot, don't, ...	count / words

Table 3.4: Linguistic markers computed by the backend

3.9.2 Speech features

The saved speech model expects a training schema with 95 columns comprising MFCC statistics, pitch, energy, spectral descriptors, chroma, duration and pause features. The inference code extracts MFCC statistics and deltas, chroma, spectral 11

contrast and zero-crossing features and aligns the feature values according to the saved feature-column names, with zeroes entered for features not present in the speech sample. The earlier baseline feature extractor extracted only 27 of the 95 features. The saved extractor has not been executed, so the speech feature contract remains unresolved. Additional prosodic features are used for the speech display summary.

3.9.3 Behavioural features

The behavioural Random Forest model expects 50 input features. The API collects 7 behavioural features (sleep hours, sleep quality, activity level, social interaction, stress level, exercise minutes and mood rating). These seven features are entered into the first seven positions of the input feature vector, with the remaining 43 features being entered as zeroes. The features collected by the API have not been verified to correspond with the training features for the behavioural model.

3.10 Machine-Learning Model Design

Modality	Algorithm	Configuration found	Output
Text	Logistic regression on TF-IDF	C = 2.0, balanced class weights	3-class risk probabilities
Speech	XGBoost classifier	300 trees max, depth 8, learning rate 0.1, early stopping 20; 95 input features	8 emotions, mapped to 3 risk tiers
Behavioural	Random Forest	100 trees, depth 5, balanced class weights; 50 input features	3-class risk probabilities
Fusion	XGBoost meta-classifier	150 trees, depth 4, learning rate 0.05; 9 inputs	3-class risk probabilities

Table 3.5: Models actually present in the repository

The folder holding the text model is named text_mentalbert, and some repository documents mention MentalBERT or Bio_ClinicalBERT. The loaded artefacts are a TF-IDF vectoriser and a scikit-learn logistic regression; no transformer model is implemented or used.

3.11 Multimodal Integration

Multimodal fusion is implemented at inference time. Each available modality yields a probability vector over (Low, Moderate, High); missing modalities are represented by a uniform vector (1/3, 1/3, 1/3). The three vectors are concatenated into a nine-element vector p and passed to the XGBoost meta-classifier. If the meta-classifier were unavailable the code would fall back to a simple average.

$$x = [p_text, p_speech, p_behav] \text{ in } R^9 \quad (3.2)$$

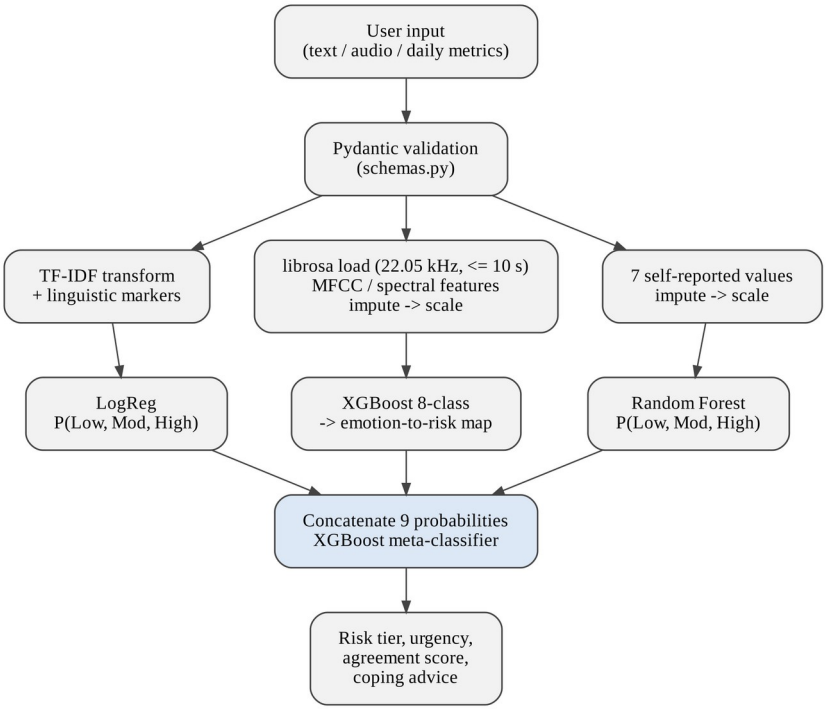


Figure 3.2: Data-processing and late-fusion pipeline

Importantly, the meta-classifier was trained on synthetic probability vectors sampled from Dirichlet distributions conditioned on randomly chosen labels (Section 5.8). It has therefore never seen real joint outputs of the three base models.

3.12 Risk-Classification Methodology

Speech emotion probabilities are converted to risk probabilities by summing probabilities within the emotion-to-risk map in Table 3.6. This mapping is a design choice of the project, not a clinically validated one.

Emotions	Risk tier
neutral, calm, happy, surprised	Low
fearful	Moderate
angry, sad, disgust	High

Table 3.6: Emotion-to-risk mapping used for speech

Cross-modal agreement A is the fraction of modality pairs that predict the same tier (A = 1 if only one modality is present). With final high-risk probability pH, the policy is: CRITICAL if pH >= 0.6; WARNING (escalate) if pH >= 0.35; and a low-confidence flag, with WARNING and escalation, if the fused confidence is below 0.5 or A is below 0.5.

$$A = (\text{number of agreeing modality pairs}) / (\text{total pairs}) \quad (3.3)$$

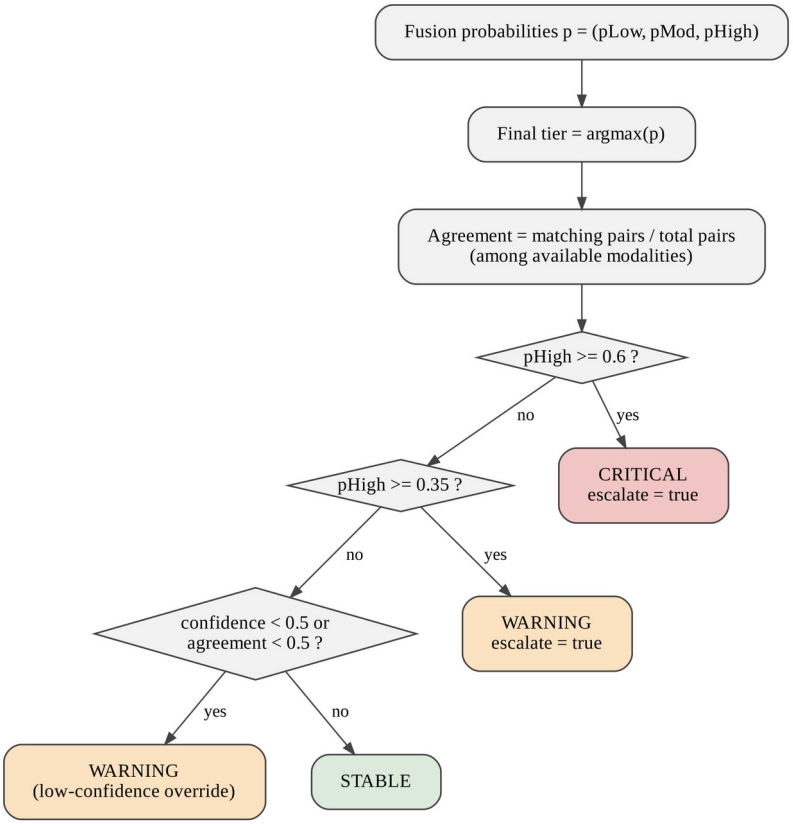


Figure 3.3: Escalation and urgency decision flow

3.12.1 PHQ-9 and GAD-7 scoring

The assessment code uses the PHQ-9 and GAD-7 scoring bands and implements a multi-stage flow for the assessments. The flow includes obtaining consent, going through the items one-by-one, reviewing the results, asking contextual questions, issuing safety disclosures if indicated, and presenting the results, symptom breakdown and next-step recommendations. The PHQ-9 question 9 is used as a safety signal, with direct disclosures triggering critical urgency and crisis resource recommendations. The scoring bands are used as a screening aid, and not for diagnosis.

3.13 Backend Architecture

The backend operates through a FastAPI application [22] which uses a lifespan hook to load model artefacts while setting up persistence. The Pydantic schemas perform validation on request and response data while the API documentation lists 13 routes which include operations for prediction and questionnaire and chat and voice therapy and therapy session history. The final code base enables assessment orchestration through enhanced session management and conversation intent handling which stores data in databases by saving assessment attempts and sessions and messages and summaries. Users can access PostgreSQL through configuration settings although SQLite continues to serve as the default local database.

The session manager operates through defined conversational stages which include opening/check-in and exploration and deepening and collaborative direction and intervention and closing and it identifies intents for greetings and ambiguous replies and corrections and topic changes and venting and exercise and advice requests and closure and crisis disclosures. The source code contains records of these paths which determine their operational effectiveness through regression testing.

3.14 Frontend Architecture

The frontend operates through Next.js 16 App Router [23] while it uses React 19 and TypeScript and Tailwind CSS. Eight top-level routes cover the dashboard, assessment, daily check-in, journal, voice check-in, insights, chat therapy and voice therapy. The final merge transforms the assessment route into separate components which include consent and question-card and review and follow-up and safety-modal and results and it enhances chat/voice therapy interfaces and API types. The system keeps user answers available when users switch between different assessment views. Supabase remains an external authentication integration. The screenshots in Chapter 4 were captured before the final merge and do not show the new multi-stage assessment UI.

3.15 System Data Flow

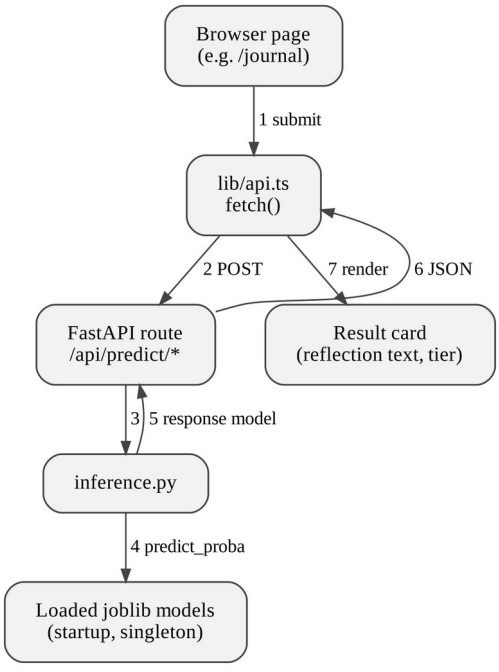


Figure 3.4: Request-response flow between browser, API and models

3.16 Algorithm or Workflow Design

```
def predict_multimodal(text, audio, behavioral):
    p_t = p_s = p_b = [1/3, 1/3, 1/3]            # uniform prior
    if text:       p_t = text_model(text);       tiers.append(argmax(p_t))
    if audio:      p_s = speech_model(audio);    tiers.append(argmax(p_s))
    if behavioral: p_b = behav_model(behavioral); tiers.append(argmax(p_b))
    p = meta_classifier(concat(p_t, p_s, p_b))
    A = pairwise_agreement(tiers)
    urgency = CRITICAL if p[High] >= 0.6 else WARNING if p[High] >= 0.35 else
STABLE
    if (max(p) < 0.5 or A < 0.5) and urgency == STABLE: urgency = WARNING
    return tier=argmax(p), urgency, A, coping_advice(...)
```

The therapy module applies message classification to all incoming messages before it decides to use an outside language-model service. The system provides acute-crisis routing through explicit paths and delivers deterministic responses and resources and supports session-based intent tracking and includes tests which verify its response to direct suicidal statements. The assessment process follows a fixed path which leads PHQ-9 item 9 and safety disclosures to their required escalation procedures. The source code contains these safeguards, but their execution status remains unverified.

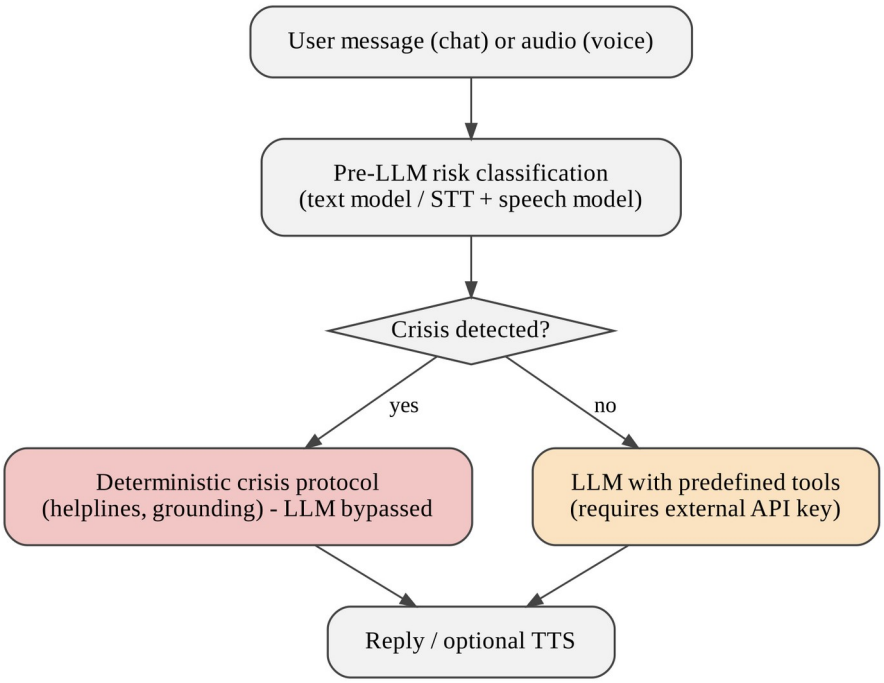


Figure 3.5: Source-level therapy safety path (crisis regression tests not yet executed)

3.17 Design Decisions and Trade-Offs

- **Classical models:** provide quick results through local operations which users can monitor yet they produce less accurate results when dealing with complex text.
- **Deterministic coping library:** prevents clinical advice generation through its operations but it produces a limited range of output results.

- **Late fusion:** enables operations with incomplete modality data but its success depends on the precision of base-model probability outputs.
- **SQLite default:** zero configuration, but not suitable for multi-user production use.

3.18 Proposed Design (Synopsis) versus Latest Implemented Repository

The synopsis described a target architecture rather than a binding inventory of completed features. Table 3.7 compares the intended design with the final source-level implementation.

Component	Proposed / earlier description	Final implementation (cf39c66)	Status
Text model	BERT / MentalBERT, possibly trained with emotion/stress datasets	TF-IDF (1-3 grams, 10,000-term cap) + Logistic Regression	Verified from artefact; High-tier recall low
Speech model	CNN-LSTM / MFCC deep model	XGBoost over saved 95-column acoustic schema; named-feature alignment	Alignment untested; zero-filled names a risk
Behavioural model	RF/XGBoost on behavioural time series	Random Forest expects 50 features; API gathers 7 fields and zero-pads	Mismatch unresolved
Fusion	Multimodal ensemble / late fusion	XGBoost meta-classifier over nine probabilities; trained on synthetic vectors	Not validated on real outputs
Assessment	PHQ-9 / GAD-7 scoring	Consent, question cards, review, follow-up, safety disclosure, score bands, domain breakdown, recommendations	Tests added; not executed
Therapy	Supportive chat/voice with crisis handling	Intent-aware session manager, session stages, crisis routing, LLM/STT/TTS clients, persistence	Crisis tests not run
Frontend	Streamlit in synopsis	Next.js 16 / React 19 / TypeScript / Tailwind; eight top-level routes	New assessment/chat/voice UI not browser-tested
Storage and deployment	PostgreSQL, Docker, cloud deployment	SQLite default; PostgreSQL optional; Docker files present	PostgreSQL and Docker untested
Tests	No detailed test strategy in synopsis	tests/test_assessment.py and tests/test_therapy.py	Not executed
Safety	High-risk escalation and emergency support	PHQ-9 item-9 and direct-disclosure safeguards; therapy crisis path and regression test	Not resolved until tests pass

Table 3.7: Synopsis proposal compared with latest implementation

CHAPTER 4: IMPLEMENTATION AND SOFTWARE DEVELOPMENT

4.1 Technology Stack

Layer	Technology	Version found
Frontend	Next.js (App Router, Turbopack), React, TypeScript, Tailwind CSS 4	Next 16.3.0, React 19.2.8
Authentication client	@supabase/supabase-js	^2.112.3
Backend	FastAPI, Uvicorn, Pydantic	FastAPI 0.160.0 (installed)
ML libraries	scikit-learn, XGBoost, joblib, NumPy, pandas	scikit-learn 1.9.1, XGBoost 3.4.1 (installed)
Audio	librosa, soundfile	librosa 1.0.0 (installed)
Persistence	SQLite (default), PostgreSQL via psycopg2 (optional)	-
External (optional)	LLM providers, speech-to-text, ElevenLabs text-to-speech, Supabase	API keys required

Table 4.1: Technology stack verified from configuration files

4.2 Repository Structure

Path	Contents	Used by running app?
backend/app/	FastAPI app, inference, schemas, assessment, therapy, DB, engine clients	Yes
frontend/src/	Next.js pages, components, API client, auth context	Yes
models/	Four trained artefact groups with metrics and plots	Yes (loaded at startup)
notebooks/	Six EDA notebooks and three modelling notebooks	No (training and analysis)
src/	Python package skeleton; all files are empty __init__.py	No
tests/, configs/, scripts/	Placeholder files at the baseline audit (setup_env.ps1 script); tests/ now holds the two new test modules	No
data/	Two SQLite databases; raw and processed data are absent	SQLite file used
docker/	Dockerfiles and compose file	Not verified
reports/	Ablation CSV/JSON and figures	No

Table 4.2: Repository directories and their actual contents

4.3 Backend Implementation

main.py creates the FastAPI app, loads environment variables, registers CORS for the local frontend and defines the routes. model_loader.py defines a singleton Models object that loads each joblib artefact inside a try/except block, so a missing model sets a flag to false without crashing start-up. /health returns these flags. schemas.py defines 18 Pydantic request and response models, for example a text request that requires at least one character.

Pydantic request and response models, for example a text request that requires at least one character.

```
@asynccontextmanager
async def lifespan(app: FastAPI):
    models.load_all(MODELS_ROOT)   # text, speech, behavioral, fusion
    init_db()                      # SQLite fallback unless DATABASE_URL set
    yield
```

4.4 Frontend Implementation

Each page is a client component that calls a typed helper in `lib/api.ts` (predictText, predictSpeech, predictBehavioral, predictMultimodal, submitAssessment, startSession, sendChatMessage, sendVoiceTherapy, and others). The base URL is read from NEXT_PUBLIC_API_URL and defaults to http://localhost:8000. The journal page, for instance, calls predictText and translates the returned risk tier into gentle wording.

4.5 Dataset Loading and Preprocessing Implementation

Preprocessing is performed in notebooks, not in reusable modules: the src/ package is empty. The notebooks read processed CSV files that are not distributed with the repository, so preprocessing could not be re-executed. This is recorded as a limitation.

4.6 Model Implementation

Training code exists in 01_text_modeling.ipynb, 02_speech_modeling.ipynb and 03_multimodal_fusion.ipynb. The speech notebook trains an XGBoost model and a Random Forest and keeps the one with the higher macro F1 (0.4807 versus 0.4703). The fusion notebook also trains the behavioural Random Forest. Inference code (Section 3.9) loads the saved artefacts.

4.7 API Endpoints and Communication

Method	Path	Purpose
GET	/health	Server and model-load status
POST	/api/predict/text	Text risk tier, probabilities, linguistic markers
POST	/api/predict/speech	Audio (multipart field audio): emotion, risk, prosody
POST	/api/predict/behavioral	Risk tier from daily metrics
POST	/api/predict/multimodal	Late fusion, urgency, agreement, coping advice
GET	/api/assessment/questions/{test_type}	PHQ-9 or GAD-7 questions
POST	/api/assessment/submit	Score and store an assessment
POST	/api/therapy/session/start	Start chat or voice session
POST	/api/therapy/session/end	End session, store summary
GET	/api/therapy/sessions/{user_id}	List sessions
GET	/api/therapy/session/{session_id}/messages	Session messages

Method	Path	Purpose
POST	/api/chat/message	Chat therapy turn
POST	/api/voice-therapy/interact	Voice therapy turn (needs external providers)

Table 4.3: REST endpoints exposed by the backend (13 verified in the OpenAPI schema)

4.8 User Interface Implementation

Figures 4.1–4.8 are genuine screenshots captured from the baseline application using a headless browser. The sandbox blocked Google Fonts, so a local Material Symbols font was supplied and the text font fell back to a default serif face during capture. These screenshots pre-date commit cf39c66; in particular, Figure 4.2 does not show the final multi-stage assessment flow.

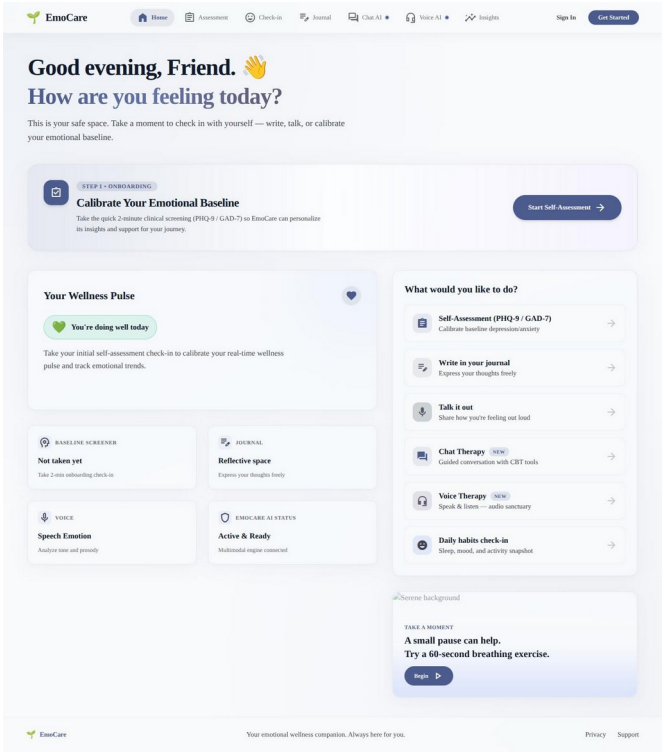


Figure 4.1: Home dashboard of the EmoCare web application

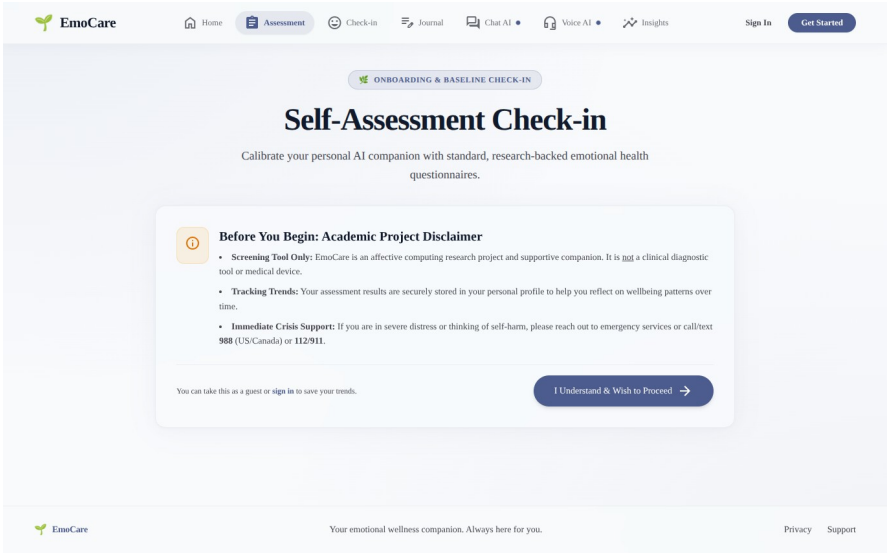


Figure 4.2: Self-assessment page (PHQ-9 and GAD-7 entry point)

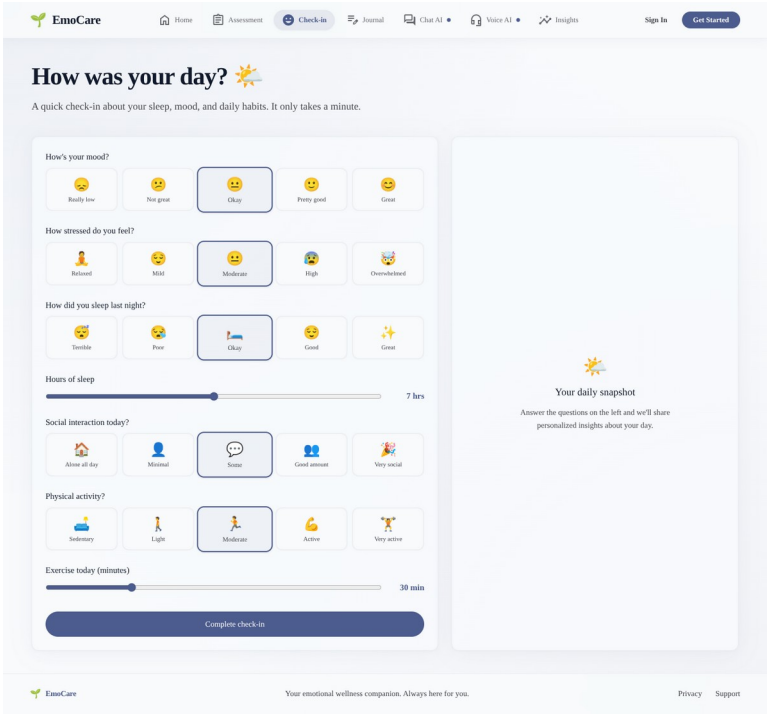


Figure 4.3: Daily check-in page for behavioural metrics

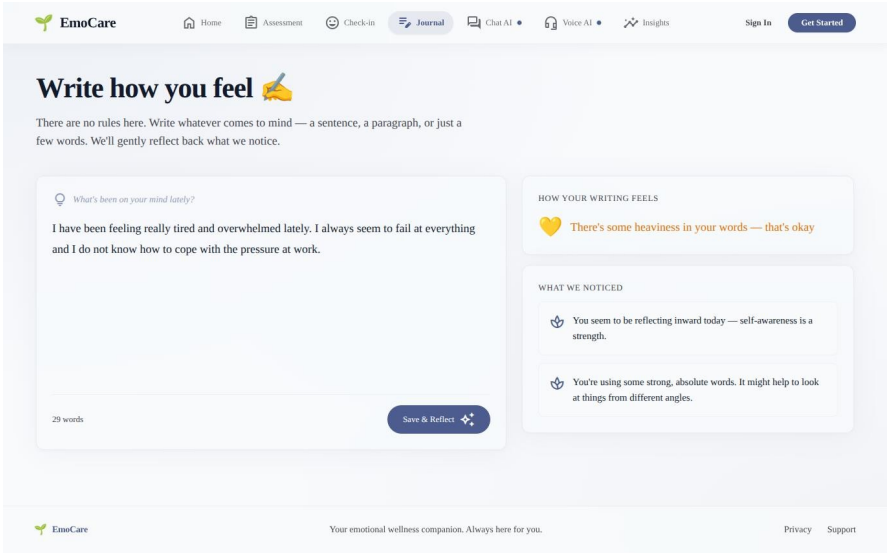


Figure 4.4: Journal page after submitting an entry: the reflection panel shows the model-driven feedback

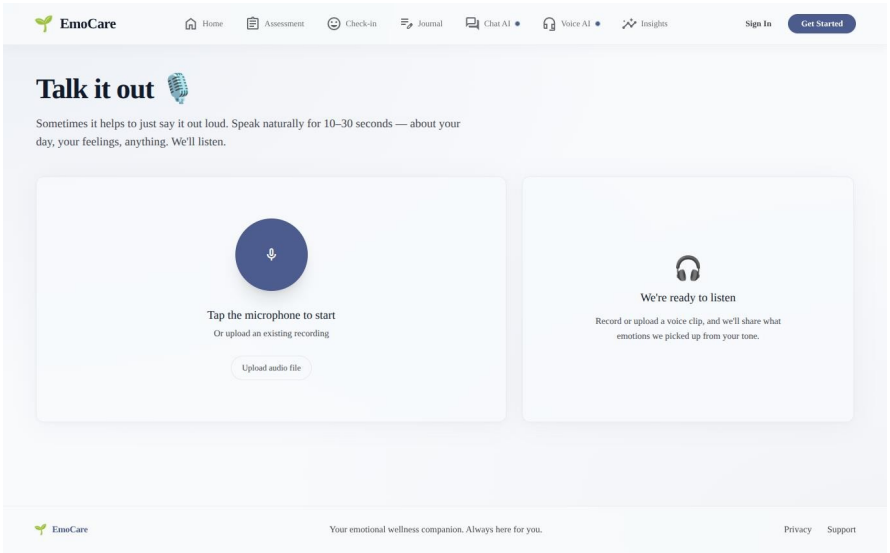


Figure 4.5: Voice check-in page

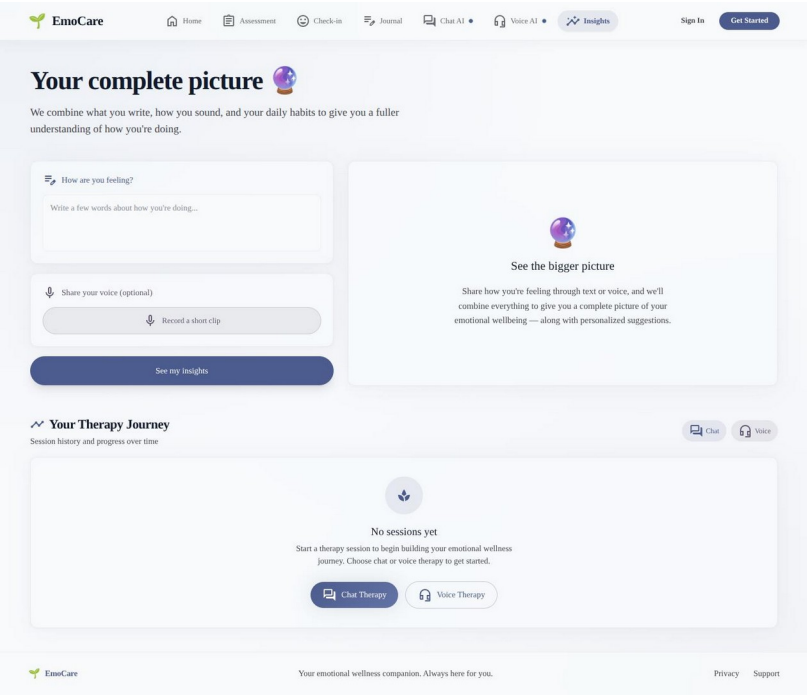


Figure 4.6: Insights page

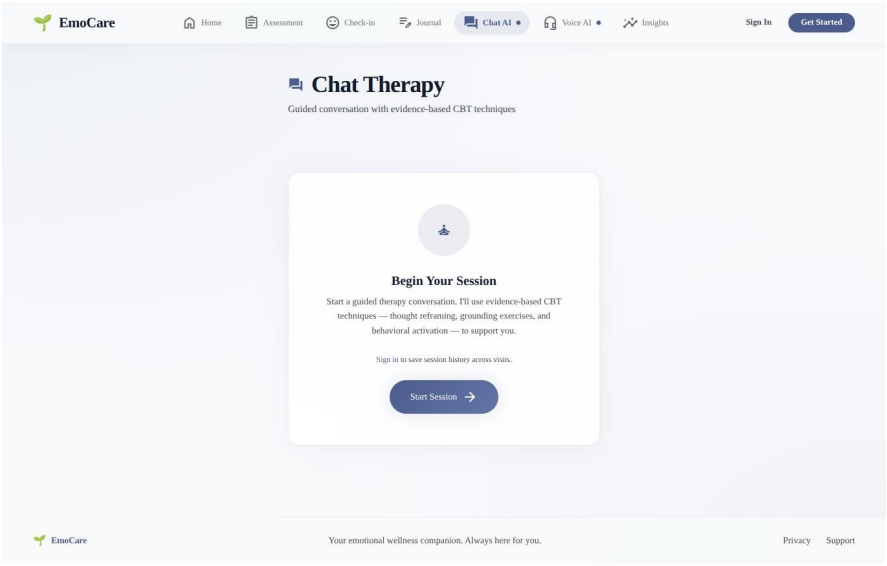


Figure 4.7: Chat therapy page

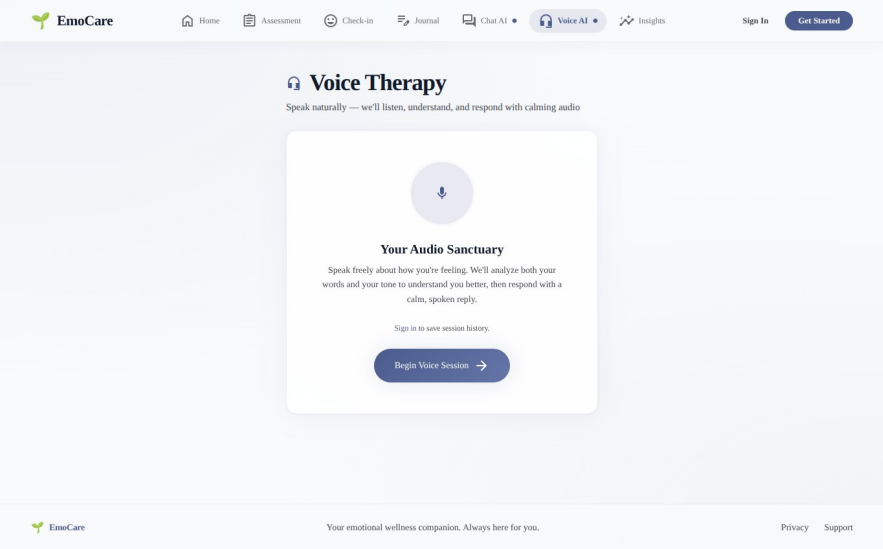


Figure 4.8: Voice therapy page

4.9 Model Inference Workflow

For a journal entry the browser posts JSON to /api/predict/text; the backend vectorises the text, obtains class probabilities from the logistic-regression model, selects the most probable tier, computes the three linguistic markers and returns them. For audio, the file is written to a temporary file, features are computed with librosa, imputed and scaled with the saved transformers, classified by XGBoost, and mapped to a risk tier. Fusion then combines available outputs as described in Section 3.11.

4.10 Application Configuration

Variable	Used by	Needed to start?
NEXT_PUBLIC_SUPABASE_URL	Frontend Supabase client	Frontend route rendering needs a value (placeholder used in testing)
NEXT_PUBLIC_SUPABASE_ANON_KEY	Frontend Supabase client	Routes returned HTTP 500 when missing; a placeholder rendered them
NEXT_PUBLIC_API_URL	Frontend API client	No (defaults to http://localhost:8000)
DATABASE_URL	Backend persistence	No (SQLite fallback)
LLM_PROVIDER and provider API keys	Chat/voice therapy	Only for LLM replies
ELEVENLABS_API_KEY, ELEVENLABS_VOICE_ID	Voice synthesis	Only for spoken replies

Table 4.4: Configuration items (values withheld)

4.11 Installation and Execution Procedure

The following commands were executed successfully in the verification environment.

```
git clone https://github.com/Dhruvone8/EmoCare && cd EmoCare
python3 -m venv venv && . venv/bin/activate
pip install -r backend/requirements.txt python-dotenv
cd backend && uvicorn app.main:app --port 8000      # API on :8000
# second terminal
cd frontend && npm install && npm run dev -- -p 3000   # UI on :3000
```

A frontend environment file with the Supabase variables is required for the pages to render. Without real credentials, only placeholders can be used, and sign-in will not function.

4.12 Security, Privacy and Responsible-AI Considerations

- **Consent and privacy:** journal text and audio are sensitive. Users must provide explicit authorization to share their data because the system stores information locally until external LLM and speech and authentication services are activated.
- **Secrets:** keys are intended to be read from environment variables and .env* files are git-ignored. A hard-coded Supabase URL fallback was present in the baseline source; the current configuration must be audited before publishing.
- The speech models which were trained with acted speech and social-media text data cannot perform well with different accents and languages and cultural backgrounds and gender identities.
- **Uncertainty:** outputs are probabilities from weak classifiers; they must not be presented as findings about a person.
- **Not diagnostic:** the system must always be presented as a reflection and screening aid. Human review and professional help are the appropriate responses to concern.

CHAPTER 5: TESTING, RESULTS AND DISCUSSION

5.1 Testing Objectives

The objectives were to verify that the repository installs and runs, to exercise each API function with valid and invalid inputs, to confirm that the user interface communicates with the backend, and to examine the evidence behind the model-performance claims.

5.2 Testing Environment

Item	Value
Operating system	Ubuntu 24 (Linux)
Python / Node / npm	3.12.3 / 22.22.2 / 10.9.7
Hardware	1 CPU core, 3.9 GB RAM, no GPU
Browser	Headless Chromium (npm package @sparticuz/chromium) driven by Puppeteer
Test date	22 October 2026
Network	Restricted: Google Fonts and CDNs unreachable

Table 5.1: Testing environment

5.3 Installation and Startup Verification

Result (commit f0532e6, 22 October 2026): the Python backend started and all four model modules loaded; the Next.js development server started. All eight frontend routes initially returned HTTP 500 without the Supabase anonymous key; they returned HTTP 200 after a local, git-ignored .env.local with placeholder values was supplied. Real authentication was not tested. A fresh checkout and execution of the final commit cf39c66 was blocked by a DNS/network resolution failure.

5.4 Functional Testing

Table 5.2 records the manual tests performed on commit f0532e6 (22 October 2026). The test files added by cf39c66 were reviewed at source level only.

ID	Function	Input	Expected	Actual	Status	Remarks
TC01	Backend start-up	Run uvicorn from backend/	All 4 model modules load	4/4 loaded; version-mismatch warnings from scikit-learn (1.9.0 vs 1.9.1) and XGBoost pickle	Pass	Warnings recorded
TC02	GET /health	After start-up	HTTP 200, all flags true	status ok; text, speech, behavioral, fusion all true	Pass	
TC03	Text, positive passage	Short upbeat sentence	Low tier	Low (p = 0.627)	Pass	
TC04	Text, distressed passage	"I feel completely hopeless... always alone"	Moderate or High	Low (pHigh = 0.293)	Fail	Consistent with 26% saved high-risk recall
TC05	Text validation	Empty string	Validation error	HTTP 422	Pass	

ID	Function	Input	Expected	Actual	Status	Remarks
TC06	Behavioural endpoint	7 poor-wellbeing values	High	High (p = 0.62)	Pass	Input mismatch, Sec. 5.11
TC07	Speech endpoint	3 s synthetic tone (not speech)	Pipeline runs	Returned "sad" (0.91), High risk	Pass (functional only)	Not an accuracy test
TC08	Multimodal fusion	Text "I feel hopeless and exhausted" plus partial behavioural inputs	Elevated or flagged	Low (0.97), STABLE, agreement 1.0	Concern	Safety observation
TC09	PHQ-9 scoring	Nine answers of 3	27, Severe, High	27, Severe, High	Pass	
TC10	PHQ-9 validation	Three answers	Rejected	HTTP 400	Pass	
TC11	Session start	user and mode chat	Session created	session_id 1 returned	Pass	
TC12	Crisis gate	"...do not want to be alive anymore"	Crisis protocol, helplines	Risk Low; crisis_activated false; breathing tip	Fail	Critical safety defect
TC13	Frontend routes (8)	Open each route	HTTP 200	500 without Supabase key; 200 for all 8 with placeholders	Pass (workaround)	Sec. 5.3
TC14	Journal end to end	Type entry, click Save & Reflect	Result panel	Reflection panel displayed (Fig. 4.4)	Pass	
TC15	TypeScript check	tsc --noEmit	No errors	No errors	Pass	
TC16	ESLint	npm run lint	No errors	3 errors, 4 warnings	Fail	See Table 5.6, item 8
TC17	Automated suite	tests/ directory	Tests present	No automated suite at f0532e6	N/A	Test modules added later in cf39c66
TC18	External services / deployment	LLM, STT, TTS, auth, mic, PostgreSQL, Docker	Successful integration	Not tested	Not tested	Need credentials/hardware
TC19	Assessment regression tests	tests/ test_assessment.py	All assertions pass	Source exists; not executed	Not tested	Questionnaire, scoring, safety signals, follow-up, persistence
TC20	Therapy regression tests	tests/test_therapy.py	All assertions pass	Source exists; not executed	Not tested	Intent, sessions, sanitisation, crisis-message response

Table 5.2: Test cases and recorded results

5.5 API Testing

The source documents 13 routes spanning health, text/speech/behavioural/multimodal prediction, assessment question/submit, therapy session start/end/list/messages, chat, and voice therapy. Responses recorded at commit f0532e6 appear in Table 5.2; the routes and the updated LLM/STT/TTS clients were not re-run against cf39c66.

5.6 Frontend Testing

Eight routes rendered in the baseline build, and Figures 4.1–4.8 are genuine screenshots from it. The journal path was exercised end to end; other pages were checked mainly for rendering. A browser run of the final build, form-submission checks, microphone checks and a console review remain unperformed.

5.7 Model Evaluation Methodology

The evaluation figures below are repository artefacts (metrics.json, confusion matrices and ablation_study.json). They could not be regenerated because the raw and processed datasets are absent. Metrics used include accuracy, precision, recall and F1-score; macro F1 averages class F1 values. The 0.744 displayed as “speech high-risk recall” is actually the recall for the sad emotion class, not recall for a high-risk tier.

5.8 Experimental Results

5.8.1 Text model (reproduced from repository artefacts)

Class	Precision	Recall	F1-score	Support
Low	0.865	0.848	0.856	429
Moderate	0.274	0.339	0.303	59
High	0.306	0.262	0.282	42
Accuracy			0.745	530
Macro average	0.481	0.483	0.481	530

Table 5.3: Text risk classifier: saved test metrics (530 test samples)

Accuracy is dominated by the majority Low class. Recall for the High class is only 26.2%, meaning about three in four high-risk texts are missed.

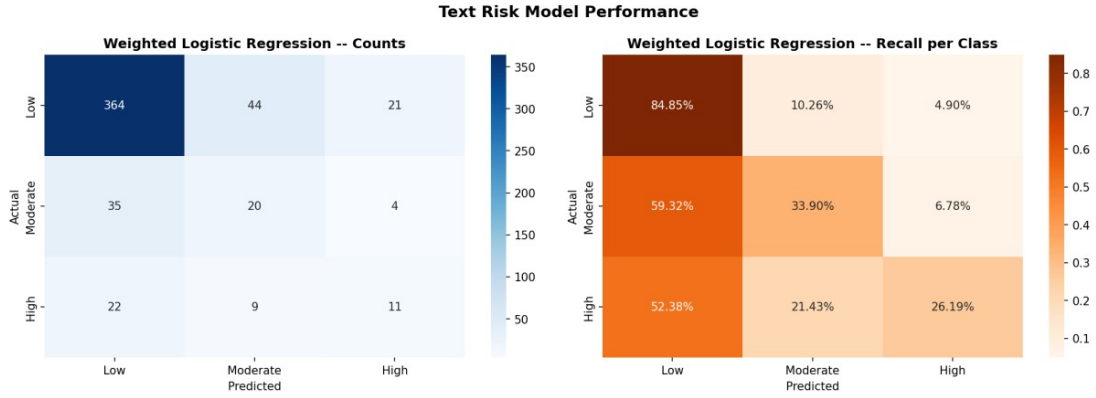


Figure 5.1: Confusion matrix of the text classifier (repository artefact)

5.8.2 Speech emotion model (reproduced from repository artefacts)

Emotion	Precision	Recall	F1-score	Support
angry	0.741	0.628	0.680	164
calm	0.533	0.333	0.410	24
disgust	0.453	0.524	0.486	164
fearful	0.508	0.366	0.426	164
happy	0.529	0.390	0.449	164
neutral	0.500	0.402	0.445	132
sad	0.436	0.744	0.550	164
surprised	0.355	0.458	0.400	24
Accuracy / macro F1			0.507 / 0.481	1000

Table 5.4: Speech emotion classifier: saved test metrics (8 classes, 1,000 test samples)

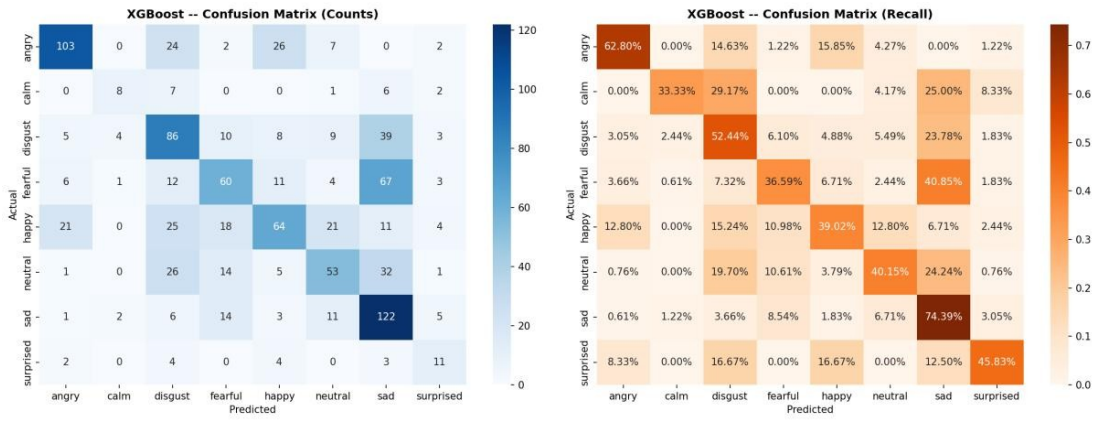


Figure 5.2: Confusion matrix of the speech emotion classifier (repository artefact)

5.8.3 Behavioural model

The behavioural notebook computes metrics by predicting on the same data that was used to fit the model, so no held-out result is available. The values 0.82 accuracy, 0.785 macro F1 and 0.85 high-risk recall appear in the ablation file, but no notebook cell that produces them was found: they are unverified.

5.8.4 Fusion engine and ablation

The Dirichlet distributions possess parameters which direct the identification process to select the specific label. The XGBoost meta-classifier receives its training and testing data from a synthetic set which contains 250 samples for testing purposes. The system shows perfect results because of its design but these numbers fail to represent actual performance in real-world situations.

Configuration	Accuracy	Macro F1	High-risk recall	Provenance
Text only	0.745	0.481	0.262	Matches text metrics.json
Speech only	0.507	0.481	0.744	Accuracy/F1 match speech metrics.json. The 0.744 equals the recall of the "sad" class, not a High-tier recall: label questionable
Behavioural only	0.82	0.785	0.85	Source not found in notebooks: unverified
Late fusion	1.00	1.00	1.00	Synthetic data: not valid evidence

Table 5.5: Ablation values in the repository and their provenance

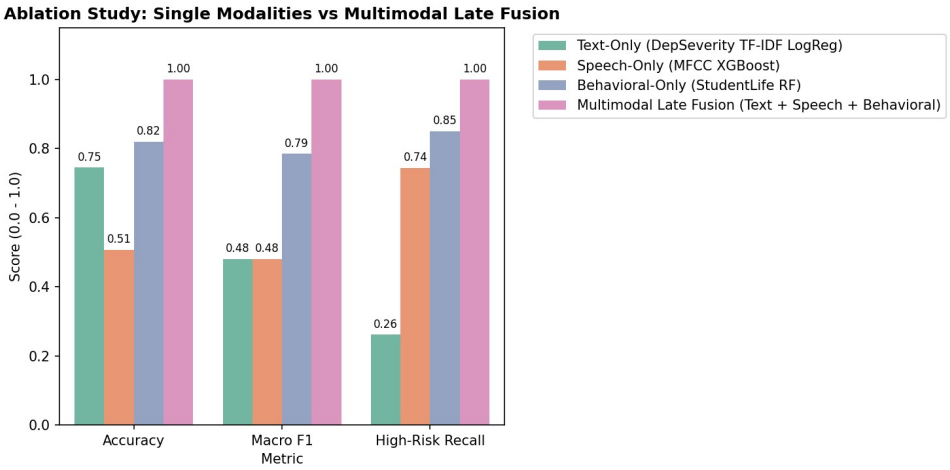


Figure 5.3: Ablation chart shipped with the repository (values discussed in Table 5.5; the fusion bar reflects synthetic data)

5.9 Comparative Results

The research team did not perform direct comparisons between their results and external published data because the external studies used different datasets and different methods to split data and define labels. The GoEmotions baseline F1 of 0.46 [5] concerns a different 28-way task and is not comparable to the three-tier risk task here.

5.10 Discussion

The repository demonstrates an integrated prototype architecture, and the final source adds structured assessment and conversation/session flows. Text and speech artefacts show macro F1 values near 0.48, and High-tier recall for text is 0.262. The new test files specify regression checks, including crisis handling, but no pass results are available. Claims of multimodal improvement remain unsupported because the fusion meta-classifier was trained on synthetic probability vectors.

5.11 Limitations and Failure Cases

Table 5.6 lists the ten limitations of the final version (commit cf39c66).

No.	Finding	Evidence	Impact
1	Crisis-safety handling and new tests not verified	PHQ-9 item-9, direct-disclosure and therapy crisis safeguards exist in source; tests/test_assessment.py and tests/test_therapy.py have not been executed	Safety blocker: not suitable for real users until the tests and explicit crisis-message scenarios pass
2	Speech feature schema alignment unverified	Saved model expects 95 columns; the extractor aligns by name and zero-fills missing names; no schema comparison test	Speech predictions may be unreliable; compare feature names, order and dimensions
3	Behavioural input mismatch	Model expects 50 inputs; the API collects 7 self-report values and zero-pads 43	Deployed behavioural output is not validated
4	Fusion trained on synthetic data	Meta-classifier trained on synthetic probability vectors and labels	The reported 100% performance is not valid real-data evidence
5	Training data and reproducibility missing	Raw and processed datasets unavailable; source, licence and speaker-independent split unverified	Model metrics cannot be reproduced or audited
6	Low text high-risk recall	High-tier recall is 0.262 in the saved text-model metrics (macro F1 0.481)	Most high-risk texts may be missed (Table 5.3 and Section 5.8.1)
7	External integrations and deployment untested	LLM, STT, TTS, Supabase authentication, live microphone, PostgreSQL and Docker were not tested	No verified claim of end-to-end operation with external services
8	Configuration, lint and data-hygiene audit pending	scikit-learn/XGBoost version warnings; 3 ESLint errors and 4 warnings; hard-coded Supabase URL fallback not confirmed removed; tracked data/assessments.db may hold test data	Re-run lint, audit configuration, and isolate or clean the database
9	Possible silent fallback when the text model is absent	Inference code can return fixed Low-skewed probabilities if artefacts fail to load	Model-load failure must fail closed or warn clearly; test this path
10	Clinical, privacy and generalisation limits	Models and risk mapping are not clinically validated; training data is acted speech and social-media text; data may reach external providers when enabled	Outputs must not be used as findings about a person; consent, retention and crisis-resource review needed

Table 5.6: Verified defects and limitations (final version, cf39c66)

5.12 Comparison of Objectives and Outcomes

Objective	Outcome
O1 Collect inputs through a web interface	Achieved: 8 routes rendered and the journal path tested; the final UI was not re-run
O2 Extract features and train classifiers	Partly achieved: model artefacts exist; training not reproducible; speech schema alignment unverified; behavioural input mismatch remains
O3 Late fusion and escalation	Fusion code present but trained on synthetic probabilities; real-world benefit not demonstrated
O4 PHQ-9 / GAD-7 scoring	Scoring implemented with consent, review, follow-up and deterministic safety handling; new tests not run
O5 Conversation module with crisis path	Intent-aware session manager and crisis regression test added; outcome unverified
O6 Verify, evaluate, report limitations	Manual verification and metrics review completed on f0532e6; cf39c66 reviewed at source level

Table 5.7: Objectives versus verified outcomes

5.13 Summary

The prototype ran and its interfaces rendered under placeholder configuration. Model quality is inadequately validated; the speech and behavioural feature contracts remain unresolved; fusion training remains synthetic; and safety handling must pass the new crisis regression tests before any consideration of use with real users.

CHAPTER 6: CONCLUSION AND FUTURE SCOPE

6.1 Project Summary

EmoCare combines a Next.js frontend and FastAPI backend with text, speech and behavioural classifiers, a late-fusion engine, PHQ-9/GAD-7 questionnaires and a supportive chat/voice module. The final merge adds a multi-stage assessment flow, explicit safety-disclosure handling, intent-aware therapy-session management, expanded persistence and two backend test modules. The baseline build was executed and screenshots were captured; the final merge was inspected at source level.

6.2 Objectives Achieved

The prototype architecture, frontend/backend API, questionnaire scoring and late-fusion code are present, and the final merge improves assessment interaction and session structure. Reproducible training, execution of the final commit, completed automated test results and trustworthy fusion evaluation have not been demonstrated.

6.3 Major Contributions

- An integrated multimodal prototype with a documented, runnable architecture.
- A transparent fusion and escalation policy based on risk probability and agreement.
- An honest audit identifying evaluation and safety gaps, with concrete remedies.

6.4 Technical Challenges

- Frontend routes required Supabase environment placeholders to render.
- Model loading emitted scikit-learn/XGBoost version warnings.
- Missing datasets prevented retraining or re-evaluation.

6.5 Remaining Limitations (Final Version, cf39c66)

The ten limitations of Table 5.6 apply to the final version of the repository:

- **Crisis-safety handling and new tests not verified.** PHQ-9 item-9, direct-disclosure and therapy crisis safeguards exist in source, but the new assessment and therapy tests have not been executed; crisis escalation needs retesting and independent review.

- **Speech feature-schema alignment unverified.** The saved classifier expects 95 columns; the extractor aligns by name and zero-fills absent names, and no exact name, order and dimension comparison exists.
- **Behavioural input mismatch.** The model expects 50 features while the API collects seven fields and zero-pads the remainder.
- **Fusion trained on synthetic data.** The reported 1.0 accuracy, F1 and recall is not a real multimodal result.
- **Training data and reproducibility missing.** Raw and processed datasets, their sources and licences, and speaker-independent splits are unavailable.
- **Low text high-risk recall.** Saved metrics show 0.745 accuracy, 0.481 macro F1 and only 0.262 High-tier recall.
- **External integrations and deployment untested.** LLM providers, STT, ElevenLabs TTS, Supabase authentication, live microphone, PostgreSQL and Docker were not tested.
- **Configuration, lint and data-hygiene audit pending.** Library version warnings, 3 lint errors and 4 warnings, the Supabase URL fallback and the tracked data/assessments.db need review.
- **Possible silent fallback when the text model is absent.** Model-load failure must fail closed or warn clearly.
- **Clinical, privacy and generalisation limits.** Models and risk mapping are not clinically validated, probabilities are not calibrated evidence about an individual, and data may reach external providers when enabled.

6.6 Future Enhancements

These are proposed improvements and are not completed work:

- Run the test suite at cf39c66 and record all outputs; add CI, coverage thresholds and regression checks for direct and ambiguous crisis language.
- Unify speech feature extraction between training and inference, test exact feature names, order and dimensions, and eliminate silent zero-fill.
- Retrain the behavioural classifier against the seven fields actually collected, or collect and validate the full 50-feature schema.
- Retrain and calibrate fusion on real held-out outputs from independently evaluated base models; document split logic and prevent leakage.

- Restore and document datasets, sources, licences, preprocessing scripts and speaker-independent splits so results are reproducible.
- Retest Supabase auth, persistence, LLM/STT/TTS, microphone capture, PostgreSQL and Docker with controlled credentials and synthetic data; capture current-build screenshots.
- Consider transformer-based text methods and explainability only after the model and safety baseline is correct and reproducible.
- Audit secret fallbacks, consent, encryption, access controls, data retention and deletion, external-provider disclosures, and region-specific crisis-resource content.

6.7 Concluding Remarks

EmoCare is an engineering prototype that contains structured assessment and intent-aware therapy logic. Its code still requires execution, regression testing, model-feature alignment checks, reproducible evaluation and safety validation. Until those checks are completed independently, the application must not be represented as clinically validated or used to make decisions about real users.

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